

MACHINE LEARNING

ST3189

COURSEWORK



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Introduction

Machine Learning is a suite of statistical and computational techniques that allow systems to learn patterns from data and make predictions. They can be categorized into supervised & unsupervised learning. Supervised techniques such as regression and classification require labelled data to train predictive models while in unsupervised learning, a statistical model is trained on un-labeled data to identify patterns and relationships.

Three analytical tasks are addressed. First, Principal Component Analysis (PCA) and K-Means Clustering identify student risk profiles from behavioural and demographic features alone without using grades. Second, regression models predict the continuous final grade G3 under two scenarios: (A) including prior grades G1 and G2, and (B) excluding them, simulating enrolment-time prediction. Third, classification models predict pass ($G3 \geq 10$) or fail, with threshold analysis to maximise detection of at-risk students.

Research questions:

RQ1: What are the optimal student clusters and do they predict outcomes?

RQ2: Which regression model best predicts G3, and how much do G1/G2 contribute?

RQ3: Can at-risk students be accurately identified, and what decision threshold is optimal?

1. Data Description and Exploratory Analysis

1.1 Dataset and Variables

The dataset contains 33 variables per student: demographics (age, sex, address), family background (parental education Medu/Fedu), lifestyle behaviours (study time, alcohol consumption Dalc/Walc, absences, goout), and academic grades (G1, G2, G3). After label-encoding 13 binary columns, one-hot encoding 4 nominal columns, and removing rows with missing values, the combined encoded dataset contains 1,044 rows and 42 columns with no missing values

1.2 Exploratory Data Analysis

Mathematics has a 32.9% failure rate ($G3 < 10$) versus 15.4% in Portuguese — more than double — establishing Maths as the harder prediction target.

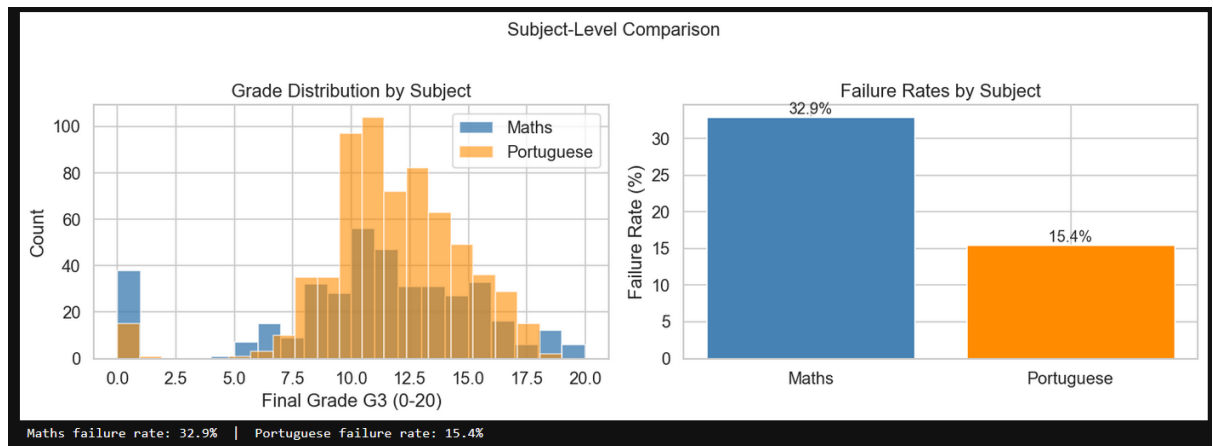


Figure 1: Math vs Portuguese failure rate

The correlation matrix over core numeric features reveals that the two periodic grades G1 ($r = 0.81$) and G2 ($r = 0.91$) are overwhelmingly the strongest predictors of G3. Among non-grade variables, the most important correlates are: number of past failures ($r = -0.38$), mother's education level Medu ($r = 0.20$), study time ($r = 0.16$), father's education Fedu ($r = 0.16$), and workday alcohol consumption Dalc ($r = -0.13$). Alcohol consumption (both daily and weekend) shows a consistent negative association with grades across the combined dataset.

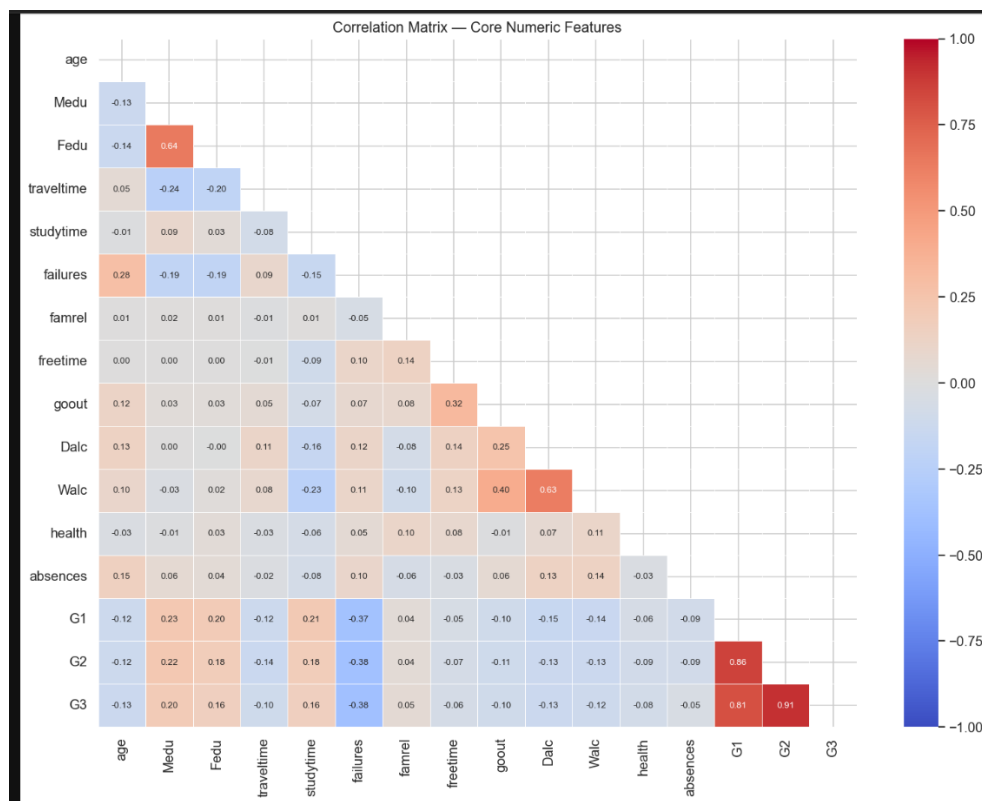


Figure 2: Correlation Matrix

Boxplot analysis confirms that median G3 rises monotonically with study time and falls steeply with past failures. Students with 3+ prior failures have a median grade below the pass threshold, identifying failures and absences as the primary actionable levers for intervention.

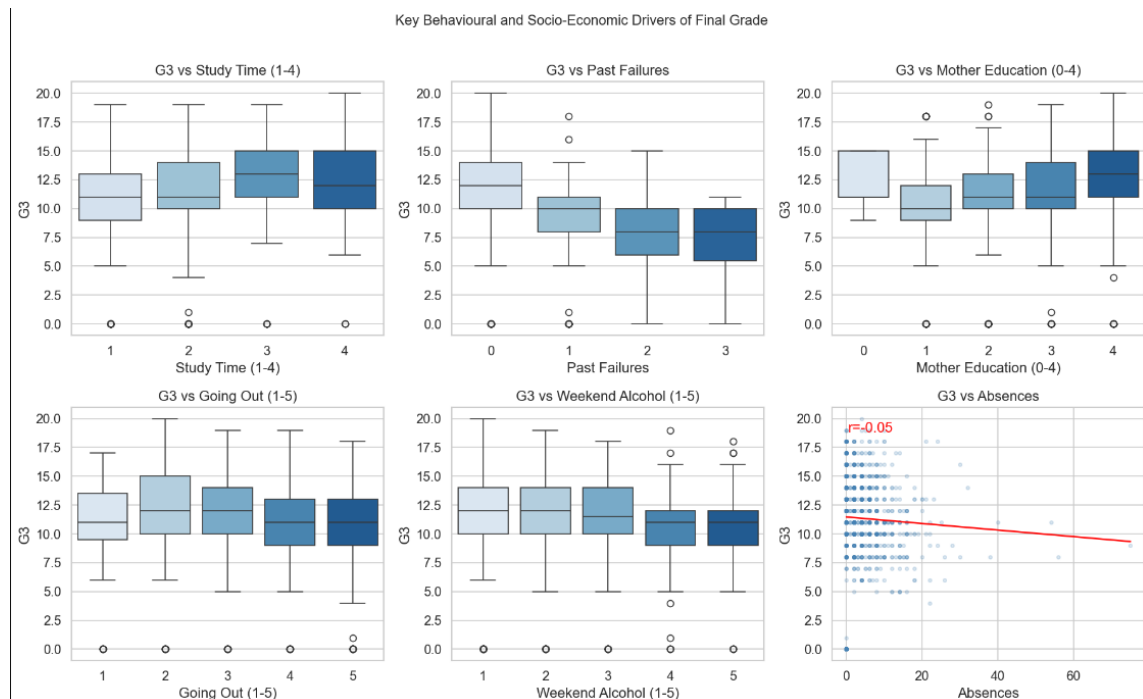


Figure 3: Boxplot a 2x3 panel of plots showing how each of six key behavioural variables (studytime, failures, Medu, goout, Walc, absences) relate to G3

Of the 1,044 combined records, 53 (5.1%) have G3 = 0. In the original dataset documentation, this value typically represents non-submission or voluntary withdrawal rather than a true grade of zero.

2. Unsupervised Learning

2.1 Methodology

Features were standardised (zero mean, unit variance) prior to analysis. PCA was applied to understand latent structure. K-Means Clustering was used with the optimal K selected via the Elbow Method and Silhouette Score. Cluster validity was confirmed with the Kruskal-Wallis non-parametric test on G3 which is a variable not used in forming the clusters. The scree plot shows that 10 components explain $\geq 90\%$ of variance, indicating a moderately high-dimensional feature space. PC1 loads heavily on Walc (0.37), Dalc (0.35), and goout (0.32), and negatively on studytime (-0.23), representing a 'risk behaviour' dimension. PC2 loads on Medu and Fedu, capturing socio-economic background.

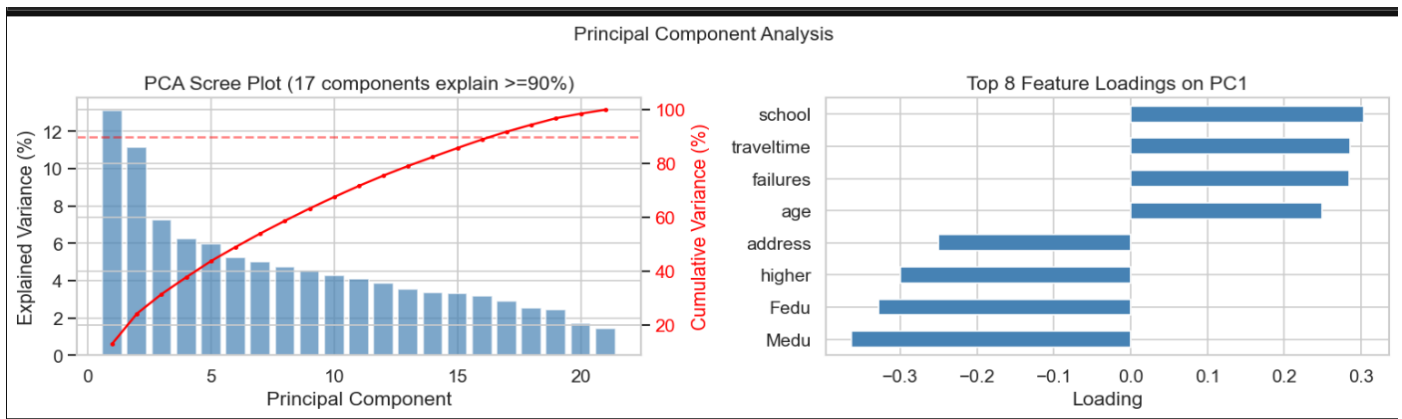


Figure 4: PCA Scree Plot and Top Feature Loadings on PC1

2.2 K-Means Clustering

Both the Elbow Method and Silhouette Score converge on $K = 2$ (Silhouette = 0.118) as the optimal number of clusters.

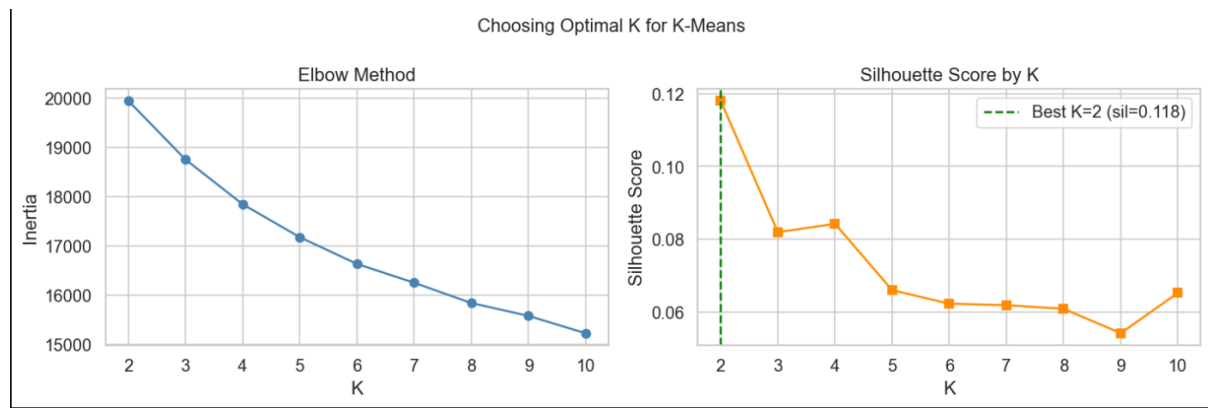
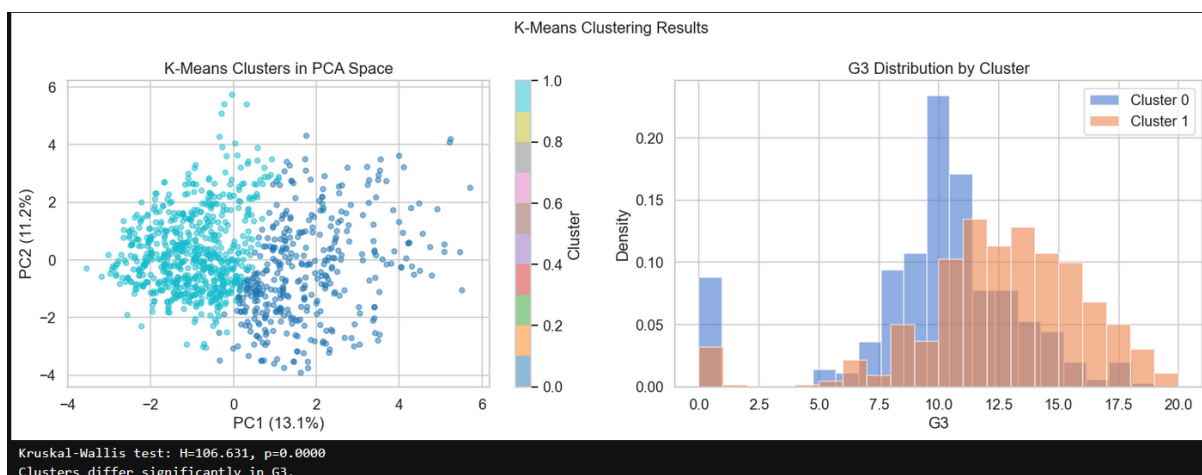


Figure 5: Elbow Method and Silhouette Score for Optimal K Selection

The K-Means algorithm (10 random initialisations, random state fixed for reproducibility) was applied with $K = 2$. A Kruskal-Wallis H-test was conducted to validate whether the resulting clusters differ significantly in G3. The result ($H = 106.63$, $p < 0.0001$) confirms that the clusters



are statistically associated with final grade, despite grades not being used in their formation which is a key validation criterion.

Figure 6: K-Means Clusters in PCA Space (left) and G3 Distribution by Cluster (right)

Cluster 0 (n=382) has lower parental education, more prior failures, higher alcohol consumption, and lower study time. Its 66.8% pass rate is 17.6 percentage points below Cluster 1, and its mean G3 (9.93) falls below the pass threshold. The two clusters represent a socio-economically disadvantaged, high-risk-behaviour profile (Cluster 0) versus an educationally supported, low-risk profile (Cluster 1). The 17.6 percentage point pass rate gap, confirmed by the Kruskal-Wallis test, validates these clusters as actionable for enrolment-time risk screening.

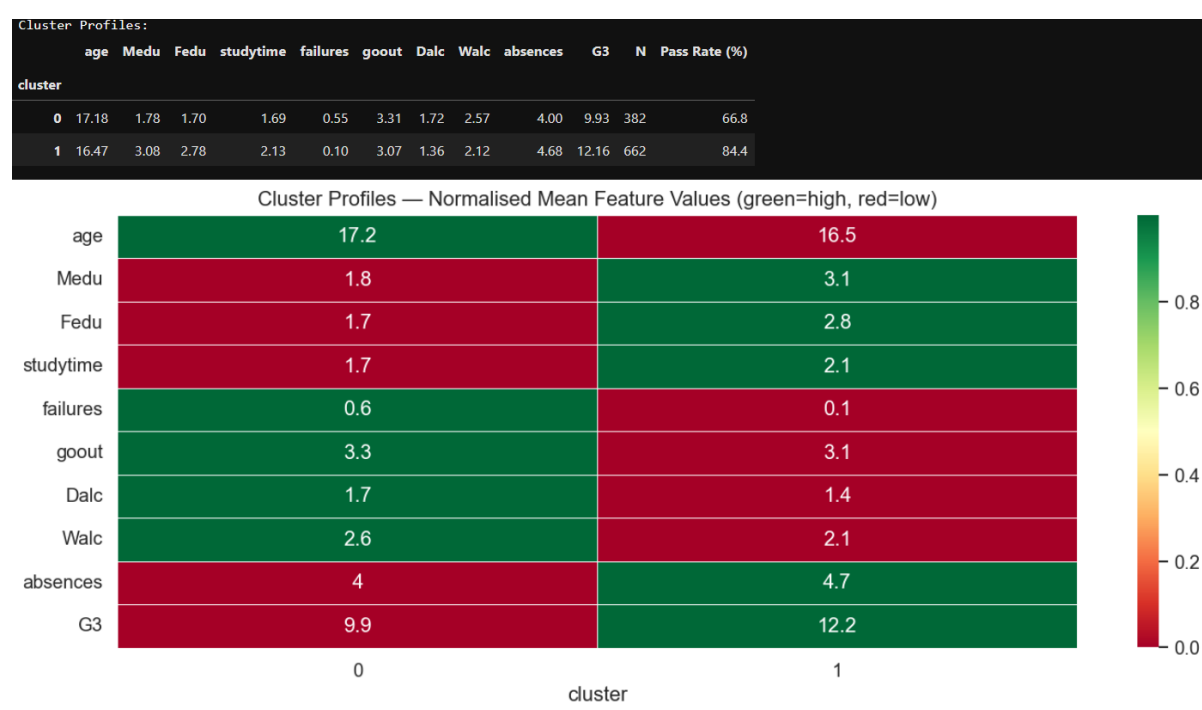


Figure 7: Normalised Cluster Profile Heatmap (green = high, red = low)

This allows us to answer **RQ1**: What are the optimal student behavioural clusters, and do they significantly predict academic outcomes?

- The optimal number of clusters is $K = 2$ (Silhouette Score = 0.118), supported by the Elbow Method.
- Cluster 0 represents an at-risk profile: lower parental education, more past failures, higher alcohol consumption, and lower study time. Mean G3 = 9.93, Pass Rate = 66.8%.
- Cluster 1 represents an on-track profile: higher parental education, fewer failures, moderate alcohol consumption, and higher study time. Mean G3 = 12.16, Pass Rate = 84.4%.
- The Kruskal-Wallis test ($H = 106.63$, $p < 0.0001$) confirms that clusters differ significantly in G3 despite grades not being used in their construction.

3. Regression: Predicting Final Grade G3

3.1 Methodology

Seven regression models were trained and evaluated under both scenarios using an 80/20 train-test split (random state fixed for reproducibility). Models included: Linear Regression, Ridge Regression ($\alpha=1$), Lasso Regression ($\alpha=0.01$), Decision Tree (max depth=6), Random Forest (200 trees), Gradient Boosting (200 estimators), and Support Vector Regression.

3.2 Analysis

With G1 and G2, the tuned Random Forest (max_depth=5, min_samples_leaf=4, n_estimators=300) achieves RMSE = 1.70 and $R^2 = 0.812$. Gradient Boosting performs similarly. Linear models achieve $R^2 \approx 0.79$, showing the G1/G2 signal is largely linear. Without G1 and G2, SVR (RBF) is the best model with RMSE = 3.397 and $R^2 = 0.254$. Linear models fail ($R^2 \approx 0.09$), confirming the remaining relationship is non-linear.

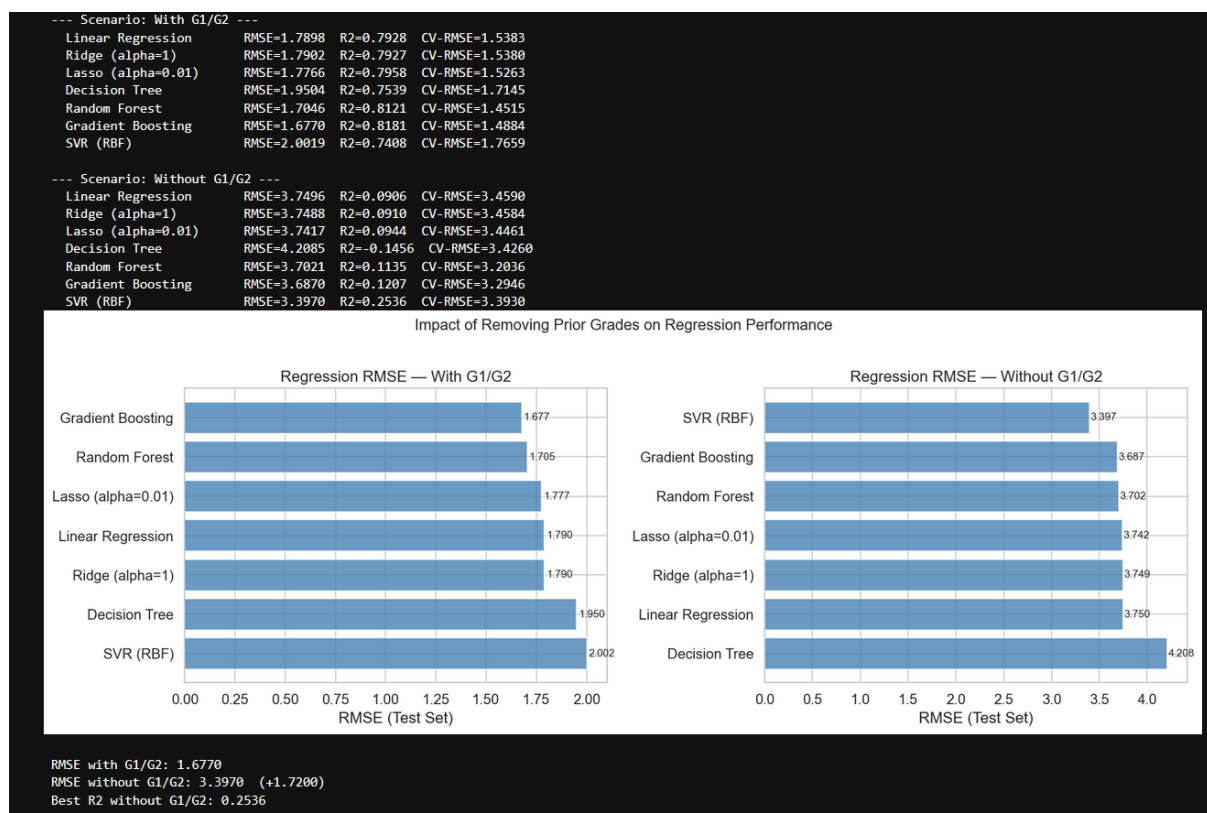


Figure 8: Regression RMSE Comparison — Scenario A (with G1/G2) vs Scenario B (without G1/G2)

Residual analysis of the tuned Random Forest (Scenario A) reveals a broadly normal residual distribution (Shapiro-Wilk $p = 0.0002$, indicating slight non-normality at the tails — expected given the integer nature of grades). The Residuals vs Fitted plot shows no

systematic pattern, confirming that the model is not systematically biased across the prediction range. The learning curve shows the cross-validation RMSE stabilising at approximately 700 training samples, indicating that the current sample size is sufficient and further data collection would yield diminishing returns.

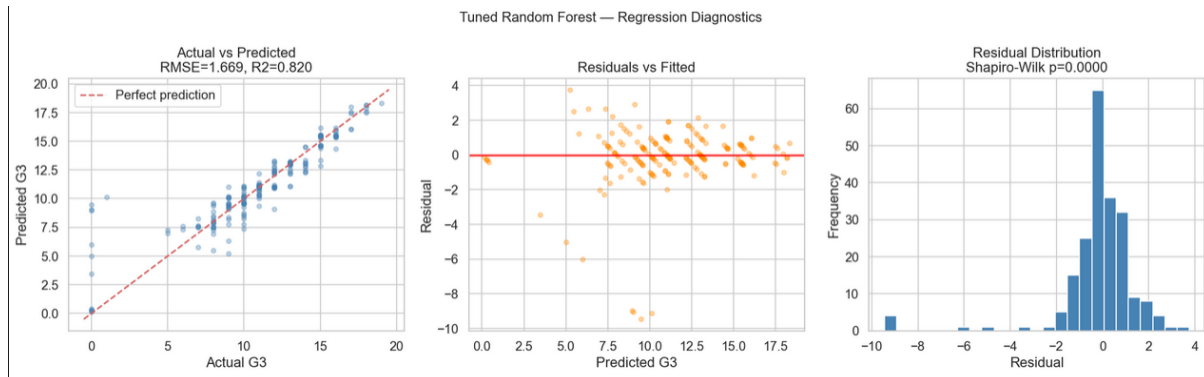


Figure 9: Tuned Random Forest Regression Diagnostics — Actual vs Predicted, Residuals vs Fitted, Residual Distribution

This allows us to answer **RQ2**: Which model best predicts final grade G3, and how much do prior grades contribute?

- Scenario A best model: Tuned Random Forest (RMSE = 1.669, $R^2 = 0.820$).
- Scenario B best model: SVR with RBF kernel (RMSE = 3.397, $R^2 = 0.254$).
- Removing G1/G2 increases RMSE by approximately 1.73 points, quantifying the information value of mid-year assessments.
- The most important non-grade features are: failures, absences, and Medu (mother's education).
- The learning curve confirms the model generalises well with no indication of heavy overfitting.

4. Classification: Predicting Pass or Fail

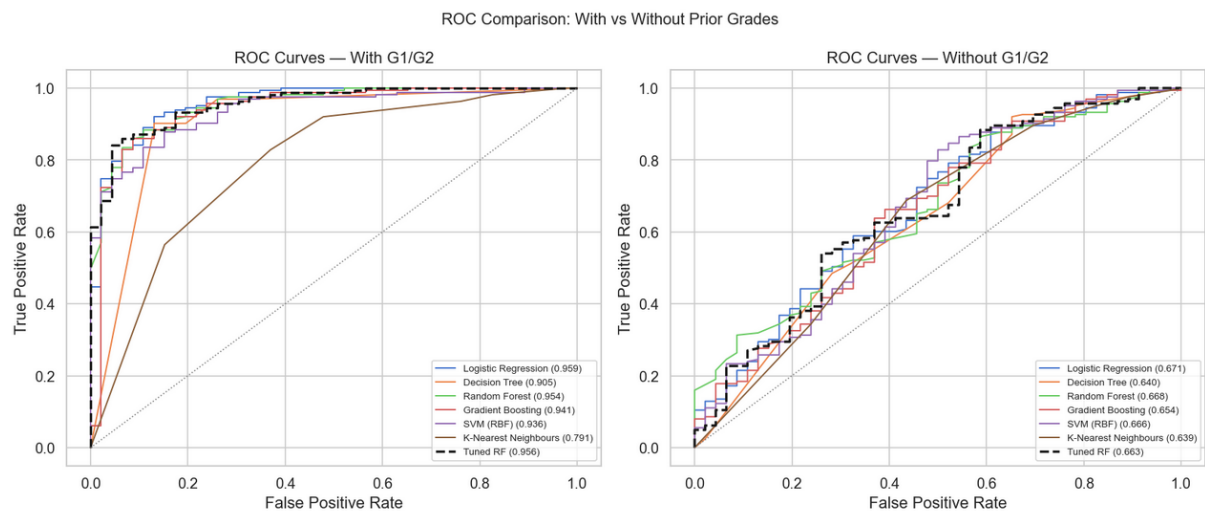
4.1 Methodology

Six classifiers were trained under both scenarios using an 80/20 stratified split: Logistic Regression, Decision Tree (depth=6), Random Forest (200 trees), Gradient Boosting, SVM (RBF), and KNN (k=7). The best model was tuned via Stratified 5-fold GridSearchCV. A threshold analysis (0.1–0.9) was performed for Scenario B. An interpretable depth-4 Decision Tree was also trained to produce human-readable rules for non-technical staff.

4.2 Analysis

Scenario A — With G1 and G2

All classifiers achieve accuracy above 89% in Scenario A, far exceeding the 78% naive baseline. The tuned Random Forest achieves accuracy = 0.900, AUC = 0.956, and F1 = 0.936. The Logistic Regression model performs comparably (AUC = 0.959), confirming that much of



the discriminative signal from G1/G2 is linearly separable. The high AUC (> 0.95) indicates excellent discrimination between pass and fail students when prior grades are available.

Figure 10: ROC Curves — All Classifiers, Scenario A (left) and Scenario B (right)

Scenario B — Without G1 and G2

Removing G1 and G2 degrades all classifiers substantially. The tuned Random Forest achieves accuracy = 0.799, AUC = 0.663, and F1 = 0.881. Importantly, the recall for the Fail class drops to 0.24 at the default threshold of 0.5 — meaning 76% of failing students are missed. This is unacceptable for a real-world early warning system and motivates threshold analysis.

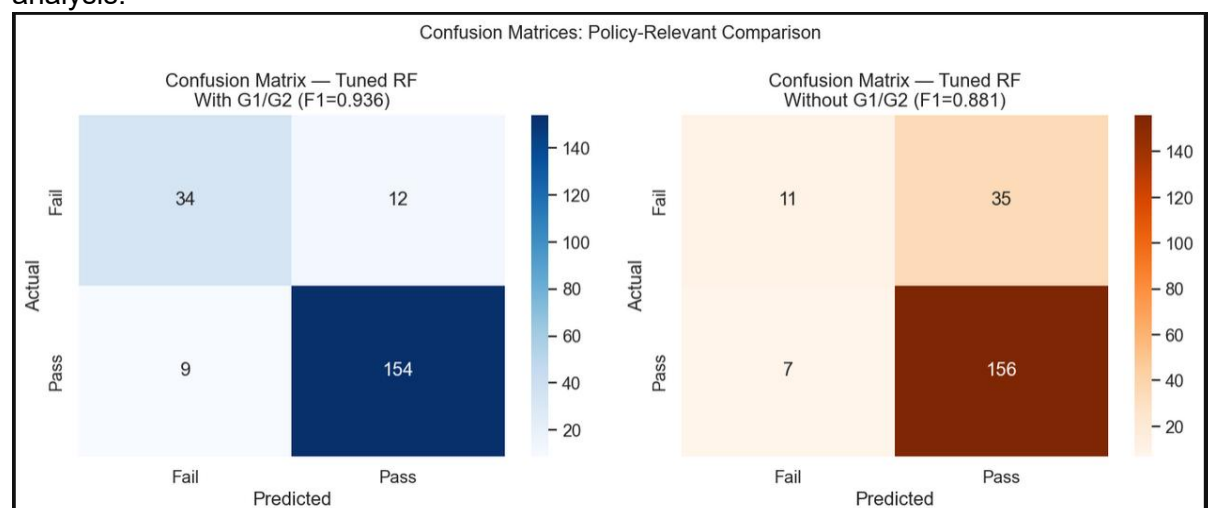


Figure 11: Confusion Matrices — Tuned Random Forest, Scenario A (left) and Scenario B (right)

To address the low recall problem in Scenario B, we evaluated precision, recall, and F1 across 80 decision thresholds from 0.1 to 0.9. The F1-optimal threshold is 0.39 (compared to the default 0.50). Lowering the threshold from 0.5 to 0.39 increases recall for the Fail class from 0.24 to approximately 0.60, at the cost of reduced precision. For schools whose primary concern is detecting at-risk students, a threshold of 0.30-0.35 is recommended: this achieves recall ≈ 0.75 while precision remains above 0.40, meaning roughly 3 in 4 failing students would be flagged and only 60% of flagged students would turn out to have passed.

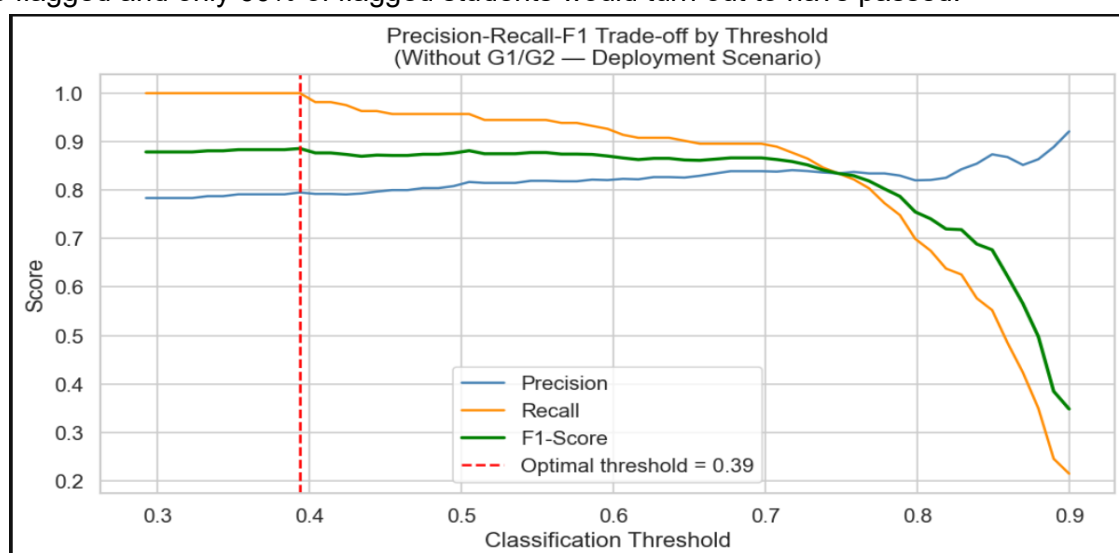


Figure 12: Precision, Recall, and F1-Score by Classification Threshold — Scenario B (Without G1/G2). Red line = F1-optimal threshold (0.39)

A depth-4 Decision Tree (no G1/G2) achieves accuracy = 0.794 with fully human-readable rules. The root split is on failures (≤ 0.5): students with no prior failures have a 95%+ predicted pass rate. This tree is the most deployable output for school counsellors who must explain decisions to parents without technical expertise.

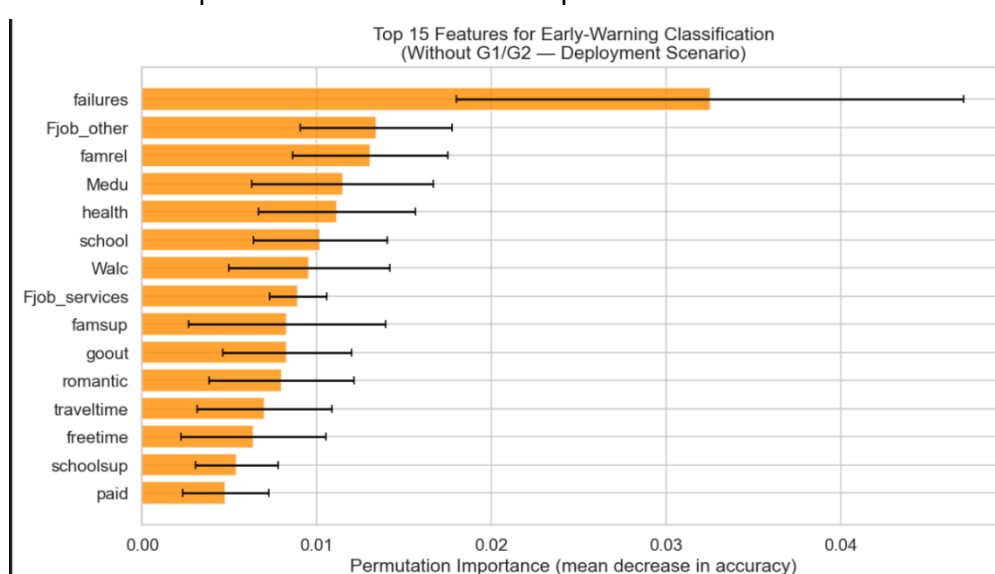


Figure 13: Permutation Importance — Tuned Random Forest, Scenario B (Without G1/G2)

5. Summary

When G1 and G2 are included, Random Forest achieves $R^2 = 0.82$ and $AUC = 0.96$ — comparable to the best results in the dataset. Without G1/G2, the best AUC falls to 0.663, but the system remains practically useful when paired with an appropriate threshold.

Unsupervised clustering produces two behavioural profiles with a statistically significant 17.6 percentage point pass rate gap, validating risk stratification at enrolment without any prior grades. For best results:

- Deploy a two-stage early-warning system: use Scenario B (threshold 0.35) at enrolment, then update with G1/G2 after the first grading period.
- Assign Cluster 0 students (pass rate 66.8%) to a support monitoring programme from day one.
- Prioritise reducing absences and supporting students with prior failures — the top actionable predictors across all three tasks.
- Use the depth-4 Decision Tree to communicate risk decisions to parents and governors transparently.

The Silhouette score of 0.118 is low, indicating overlapping clusters. The Scenario B AUC of 0.663 is modest and should not be the sole basis for high-stakes decisions. The dataset is from Portuguese schools in 2008 and may not generalise to other systems or time periods. The 5.1% of records with $G3 = 0$ (likely withdrawals) may introduce confounding in regression models.

6. Conclusion

This report applied PCA, K-Means, regression, and classification to the Student Performance dataset. The central finding is a trade-off between deployability and accuracy: models with G1/G2 are highly accurate ($AUC = 0.96$) but only usable mid-year; enrolment-time models ($AUC = 0.663$) enable earlier intervention. The most important predictors across all tasks are past failures, absences, and parental education — all known at enrolment and partially modifiable. The two-stage, threshold-adjusted early-warning system proposed here is both technically sound and accessible to non-technical school staff.

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